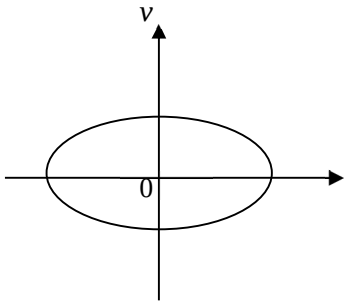


3 (a)	$a = -\omega^2 x$; { with all 3 <u>terms defined</u> correctly} $(\omega$: angular freq or ω^2 : constant of proportionality; a : acceleration; x : displacement <u>from equil position</u>)	[1]
(b)(i)	$\omega = 2\pi/T$ (where $T = 10/28$ s) = 17.6 rad s^{-1} (with correct units) (accept if T is read off graph)	[1]
(b)(ii)	At $t = 0$, x = its max positive value $\rightarrow$ eqn of motion is of the form $x = x_0 \cos \omega t$ rather than $x = x_0 \sin \omega t$ From Fig 3.2, $x_0 = 0.04$ m. $\rightarrow x = 0.04 \cos (17.6 t)$ (accept if T is read off graph) { 1 m for the cosine function; 1 m for correct values of x_0 & ω }	[1] [1]
(b)(iii)	 elliptical shape [1]	[1]
(b)(iv)	Graph showing <u>amplitude decreasing</u> with time Period remains unchanged or slightly larger.	[1] [1]
(c)(i)	$T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{0.200}{6.00}} = 1.15 \text{ s}$	[1]
(c)(ii)	Max speed $v_{\max} = \omega x_0$, where $x_0 = 0.05 \text{ m}$, $\omega = \frac{2\pi}{T}$ $= 0.273 \text{ m s}^{-1}$	[1] [1]
(c)(iii)	Total energy = $\frac{1}{2} m v_{\max}^2$ (or $\frac{1}{2} k x_0^2$) $= 7.45 \times 10^{-3} \text{ J}$	[1]
(c)(iv)	1. Using $v = \omega \sqrt{x_0^2 - x^2}$, where $x = 0.02 \text{ m}$ $= \frac{2\pi}{T} \sqrt{0.05^2 - 0.02^2}$ $= 0.250 \text{ m s}^{-1}$ (accept methods that found $t = 0.212 \text{ s}$)	[1] [1]

2. Method 1

$$\begin{aligned} E_p &= E_T - KE, \text{ where } KE = \frac{1}{2} mv^2 \text{ \& } E_T = \text{Ans in (iii)} \\ &= 7.45 \times 10^{-3} - 6.25 \times 10^{-3} \\ &= 1.2 \times 10^{-3} \text{ J} \end{aligned}$$

[1]

[1]

Method 2

$$\begin{aligned} E_p &= \frac{1}{2} m \omega^2 x^2 \\ &= \frac{1}{2} \times 0.200 \times 5.46^2 \times 0.02^2 \\ &= 1.2 \times 10^{-3} \text{ J} \end{aligned}$$